A Recurrent Linear-Optical Reservoir for Time-Series Forecasting: Matched-Capacity Benchmarks and Robustness at Measured Photonic-Hardware Noise Levels

Hüseyin Umut Işık[1,*], Arda Kara[2,*], Mehmet Alp Özaydın[1], Eren Aslan[1]

[1]Middle East Technical University, Ankara, Turkey
[2]Bilkent University, Ankara, Turkey
[*]Co-first authors; equal contribution.

July 29, 2026

Abstract

We study a linear-optical reservoir computer in which a leaky-integrated state is fed back into the phase encoding of a fixed photonic interferometer, with only a ridge readout trained. The feedback is applied classically between measurement batches, so the recurrent model consumes exactly the same shot budget as a memoryless windowed encoding and requires no optical memory or fast feed-forward. Evaluated under the standard input-driven reservoir-computing protocol, the model attains median NRMSE 0.095 on NARMA-10 and 0.241 on NARMA-20, outperforming a tuned echo state network, random Fourier features, a polynomial expansion of the input window, and a linear control, with Diebold–Mariano $p < 0.001$ against every baseline. Because the tuned configuration uses more features than the baselines, we sweep every model family across a common feature dimension: at ≈ 1400 features the photonic map attains 0.109 against 0.188 for random Fourier features and 0.268 for an echo state network, and the latter saturates beyond 2000 units while the photonic map does not. Removing the feedback, with all else fixed, doubles the error on both NARMA tasks. It ties a tuned echo state network on Mackey–Glass; loses to one on NARMA-5 — a task needing only five steps of memory, where long linear memory suffices and the photonic map's nonlinearity buys nothing; loses to random Fourier features on Lorenz-63, where per-seed errors span two orders of magnitude and five seeds cannot separate the candidates; and ranks third of six on S&P 500 realised volatility with no model statistically distinguishable from any other. Every outcome is reported unmodified. Simulating the Ascella and Belenos operating points read from Quandela's cloud API with threshold detectors, we find accuracy essentially unchanged by photon distinguishability (flat for Hong–Ou–Mandel visibility from 0.50 to 1.00) and by multi-photon emission (+0.8% for $g^{(2)}(0)$ from 0 to 0.30), while finite sampling dominates. Since n-fold coincidence rate falls as η^n in the transmittance η, this inverts the usual design instinct: photon number, not photon quality, is the binding constraint. Testing that directly, we find no detectable benefit from photon indistinguishability on seven of eight task and photon-number configurations, so we conclude that the architecture is a classical random nonlinear feature map realised in optics and that the exploited resource is Fock-space dimension rather than interference. Comparing measured classical throughput against measured device rates, no crossover with simulation exists at any photon number on current hardware; a crossover at twelve photons in twenty-four modes would require roughly a fourfold improvement in transmittance. We make no claim of quantum advantage, and all results reported here are simulated.

1 Introduction

Reservoir computing trains only a linear readout on the state of a fixed nonlinear dynamical system (Jaeger, 2001; Maass et al., 2002). Photonic implementations are attractive because a passive interferometer realises a high-dimensional nonlinear map at no energetic cost per inference, and because boson sampling supplies a feature space whose dimension grows combinatorially with photon and mode number (García-Beni et al., 2023; Nerenberg et al., 2025). Recent work has demonstrated quantum optical reservoirs on Quandela's Ascella processor (Maring et al., 2024) and applied related architectures to financial forecasting (Amanov and Azamov, 2026; Li et al., 2025).

Two questions decide whether such a system is worth building, and the literature answers neither cleanly. First, does the optical feature map do anything a cheap classical feature map does not, at matched feature dimension and against baselines tuned with equal budget? Second, does it survive the noise of an actual device — and if not, which noise source is responsible?

This paper addresses both. Our contributions:

- **A recurrent formulation whose feedback is free on hardware.** The reservoir state is fed back into the encoding phases and integrated with a leak rate (Section 3). Because the feedback is computed classically between measurement batches, each timestep remains a single circuit configuration and a single batch of samples. The recurrent model therefore costs the same number of shots as a stateless windowed encoding.
- **Matched-capacity benchmarking.** We sweep every model family across a common feature-dimension range rather than comparing tuned configurations of different size (Section 4.2).
- **An architectural prediction that the data confirms.** NARMA-N contains the cross-lag product $u_t u_{t-N+1}$, which a linear-optical map can only form if both lags occupy the same encoding. The optimal encoding window tracks the task order (Section 4.3).
- **A noise study at measured device parameters** showing that accuracy is insensitive to distinguishability and multi-photon emission but strongly limited by coincidence count, yielding a concrete feasibility threshold (Section 5).
- **A direct test of whether interference contributes**, by interpolating between distinguishable and indistinguishable photons at otherwise identical settings. It does not, on seven of eight configurations, so we characterise the architecture as a classical random nonlinear feature map realised in optics and identify Fock-space dimension as the exploited resource (Section 6).
- **A crossover analysis** against the measured cost of simulating the same map, which finds no crossover on current hardware and quantifies the transmittance a crossover would require (Section 7).

We also report, in Section 9, that an earlier windowed version of this architecture did not outperform a ridge regression on the raw input window, and why.

2 Background and related work

An echo state network maintains $x_t = (1-a)x_{t-1} + a\tanh(W_{\text{in}}[1; u_t] + Wx_{t-1})$ with W scaled to a chosen spectral radius, and fits a linear readout on x_t (Jaeger, 2001). Its capacity is characterised by linear memory capacity (Jaeger, 2001) and, more completely, by information processing capacity over an orthogonal basis of nonlinear functionals of the input history (Dambre et al., 2012).

Quantum optical reservoirs replace the recurrent tanh map with the output distribution of a linear interferometer over Fock space (García-Beni et al., 2023; Nerenberg et al., 2025). For n indistinguishable photons in m modes injected one per input mode, the amplitude for a collision-free output pattern S is the permanent of the submatrix $U[S,T]$, with T the input modes, so the accessible feature dimension is $\binom{m}{n}$. We simulate this exactly using Perceval (Heurtel et al., 2023).

The closest prior work applies photonic or Ising quantum reservoirs to financial series (Amanov and Azamov, 2026; Li et al., 2025); Branco et al. (2024) and Bienstman et al. (2025) argue that such comparisons frequently omit the classical control that would make them informative. We take that objection seriously enough to place a linear ridge on the raw input window in every table.

3 Method

3.1 Reservoir dynamics

Let $u_t \in \mathbb{R}^d$ be the driving input. Write $\tilde{u}_t$ for the concatenation of the most recent k lags of u, where k is the *encoding window*. The reservoir keeps a state $s_t \in \mathbb{R}^P$, $P = \binom{m}{n}$, and evolves as

$$e_t = g_{\text{in}} W_{\text{in}} \tilde{u}_t + g_{\text{fb}} W_{\text{fb}} (P s_{t-1} - \mathbf{1}) + b, \tag{1}$$

$$p_t = \text{BosonSample}(U(e_t)), \tag{2}$$

$$s_t = (1-\lambda)\, s_{t-1} + \lambda\, p_t. \tag{3}$$

Here e_t are phase-shifter settings in radians, $U(e_t)$ is the resulting circuit unitary, and p_t is the distribution over collision-free outputs. The matrices W_{in}, W_{fb}, the bias b and the interferometer are fixed random draws and are never trained; only a ridge readout on $[s_t, \text{input window}]$ is fitted.

Two details matter. The state is fed back as $P s_{t-1} - \mathbf{1}$, its deviation from the uniform distribution, whose entries are $O(1)$ regardless of Fock-space size; with W_{fb} drawn at variance $1/P$, the gain g_{fb} then means the same thing at every photon number, so a photon-number ablation does not silently confound expressivity with feedback strength. And the gains $g_{\text{in}}, g_{\text{fb}}$ are free parameters rather than implicit: a naive encoding that applies the raw series directly as phases uses only a small fraction of the available 2π range, which we found to be the dominant failure mode of the predecessor architecture (Section 9).

3.2 Feedback is classical and therefore free

Equation (3) is evaluated on measured probabilities, on a classical computer, between measurement batches. Each timestep is still one circuit configuration and one batch of samples. A recurrent run therefore requires exactly as many shots as a stateless one — the recurrence adds no photonic resource whatsoever. This is what makes the architecture deployable on current cloud-accessed hardware, where a closed optical feedback loop is not available.

The corresponding cost is a scheduling one: step t's phases depend on step $t-1$'s measurement, so the timesteps of a closed loop cannot be batched into a single cloud job. We return to this in Section 8.

3.3 Exact simulation

The circuits used here are layered, $U = A_d D(\phi_{d-1}) A_{d-1} \cdots A_1 D(\phi_0) A_0$, with the A_k fixed and $D(\phi)$ a diagonal layer of encoded phases. We therefore build the A_k once and evaluate each timestep as a few $m \times m$ products followed by a batch of $n \times n$ permanents. This agrees with MerLin and with Perceval's SLOS backend to 10^{-16} while being ∼310× faster per step, which is what makes a sequential recurrent loop affordable across seeds and hyperparameter searches.

3.4 Evaluation protocol

All models pass through an identical pipeline. One continuous series is split into washout / train / validation / test. The reservoir state is carried across split boundaries: the reservoir has no trained parameters, so driving it over later data is not leakage, and restarting the state at the test boundary would penalise exactly the memory under study. Every quantity fitted from data — input scaler, feature scaler, ridge penalty and readout — is estimated on training rows only.

The ridge penalty is selected on the validation slice *per model*. This matters: a single shared penalty favours whichever feature count it happens to suit, across models whose dimensions differ by an order of magnitude.

We report NRMSE, the root mean squared error divided by the standard deviation of the target, as used in the reservoir-computing literature. Inference uses Diebold–Mariano (Diebold and Mariano, 1995) with Newey–West HAC variance (Newey and West, 1987; Andrews, 1991) and the Hansen Model Confidence Set (Hansen et al., 2011) with a stationary bootstrap (Politis and Romano, 1994).

3.5 Baselines

Every baseline receives the same Optuna trial budget as the photonic model.

- **Linear window (control)**: ridge on a window of the raw drive. This is the model any claim about optical features must clear.
- **Echo state network** with input scaling, leak rate, spectral radius and sparsity all tuned. We note that omitting the input-scaling parameter — as a surprising amount of code does — costs roughly a factor of three on NARMA-10 and makes the baseline far too easy to beat.
- **Random Fourier features** (Rahimi and Recht, 2007) on the input window, the matched-dimension random nonlinear feature map.
- **Polynomial window**: explicit monomials up to degree three. NARMA targets are polynomial in the drive, so this bounds how much of any gain is simply "some nonlinearity in the window".
- **Photonic without feedback**: the identical optical map with recurrence disabled.

4 Experiments

Tasks span three distinct sources of temporal difficulty. NARMA-5, NARMA-10 and NARMA-20 (Jaeger, 2001) are driven by i.i.d. uniform input and demand nonlinear memory of a stochastic drive. Mackey–Glass at horizon 17 is delay-induced chaos. Lorenz-63 at horizon 20 is deterministic continuous-time chaos, probing how far ahead a model can forecast before sensitive dependence destroys predictability. Monthly S&P 500 realised volatility (Corsi, 2009) is the one empirical series.

Horizon choice matters for the chaotic tasks and we set it deliberately. Lorenz-63 at horizon 1 discriminates nothing — a tuned echo state network and a plain ridge both score $\sim 10^{-4}$, because at $\mathrm{d}t = 0.02$ the next sample is nearly a linear function of the current one. Past horizon 80, NRMSE exceeds 1 for every model, meaning none beats predicting the mean. Horizon 20, roughly 0.4 Lyapunov times, is the window in which the task separates models.

4.1 Forecasting accuracy

Model	NARMA-5	NARMA-10	NARMA-20	Mackey-Glass (h=17)
Photonic (recurrent)	0.0152 (0.0043)	**0.0951 (0.0168)**	**0.2409 (0.0456)**	**0.0005 (0.0005)**
Photonic (no feedback)	0.0152 (0.0043)	0.2156 (0.0322)	0.4355 (0.1280)	0.0005 (0.0005)
Echo state network	**0.0109 (0.0023)**	0.2794 (0.1108)	0.4132 (0.1119)	0.0005 (0.0002)
Random Fourier features	0.0164 (0.0023)	0.1759 (0.0538)	0.4347 (0.0821)	0.0014 (0.0016)
Polynomial window	0.0158 (0.0033)	0.1763 (0.0460)	0.4741 (0.1844)	0.0204 (0.0146)
Linear window (control)	0.3801 (0.0450)	0.4446 (0.0904)	0.4495 (0.0969)	0.6176 (0.0592)

Table 1: NRMSE, median (interquartile range) over 5 seeds, standard input-driven protocol. Lower is better; best per column in bold. Medians rather than means because the seed distribution on the chaotic tasks is heavy-tailed: one unlucky Lorenz-63 trajectory raises a model's mean to 3.5× its median and would decide the ranking on a single draw.

Baseline vs photonic	NARMA-5	NARMA-10	NARMA-20	Mackey-Glass (h=17)	Lorenz-63 (h=20)
Photonic (no feedback)	1.0000	0.0002	0.0000	1.0000	0.4788
Echo state network	0.0572	0.0000	0.0000	0.4062	0.0248
Random Fourier features	0.3226	0.0002	0.0000	0.0004	0.9599
Polynomial window	0.1876	0.0001	0.0000	0.0002	0.0034
Linear window (control)	0.0000	0.0000	0.0000	0.0000	0.0000

Table 2: Diebold–Mariano p-values with Newey–West HAC variance, each baseline against the recurrent photonic model.

The photonic reservoir wins NARMA-10 and NARMA-20 with $p < 0.001$ against every baseline. On Mackey–Glass it ties the echo state network ($p = 0.41$); the task saturates, with three models at $5{\cdot}10^{-4}$. It wins where the task demands nonlinear memory of a stochastic drive, and not elsewhere.

Three results run against it, and are worth stating explicitly. On **NARMA-5** the echo state network wins. This is the expected place to lose: the task requires only five steps of memory, precisely the regime where a large tanh reservoir's long linear memory is sufficient and the photonic map's extra nonlinearity buys nothing — the mirror image of Section 4.4. On **Lorenz-63** random Fourier features win on the median; the photonic model has the better *mean*, but that ordering inverts under a single seed and we do not claim it. Across models the per-seed NRMSE spans 0.06 to 11.0 on this task, and five seeds cannot separate the candidates. On **S&P 500 realised volatility** it ranks third of six and no model is distinguishable from any other ($p = 0.45$–0.88); we draw no conclusion from the financial task.

Removing feedback while changing nothing else roughly doubles the error on both NARMA tasks ($p < 0.001$), which is the ablation the architecture rests on.

The Model Confidence Set retains almost every model on every task. With 200–600 test points it has little power here, and we report it for completeness rather than as a discriminating test.

4.2 Matched capacity

The tuned photonic configuration uses more features than the baselines, so a direct comparison of Table 1 would confound the feature map with its size. We therefore sweep each family across a common dimension range.

Model	Feature dim.	NRMSE
Photonic (recurrent)	1381	0.1092 ± 0.0369
Photonic (recurrent)	701	0.1643 ± 0.0156
Random Fourier features	1200	0.1876 ± 0.0604
Random Fourier features	600	0.2373 ± 0.0631
Echo state network	501	0.2660 ± 0.0402
Echo state network	1001	0.2676 ± 0.0920
Polynomial window	679	0.4566 ± 0.0488

Table 3: NARMA-10, feature dimensions between 500 and 1500, mean over 3 seeds.

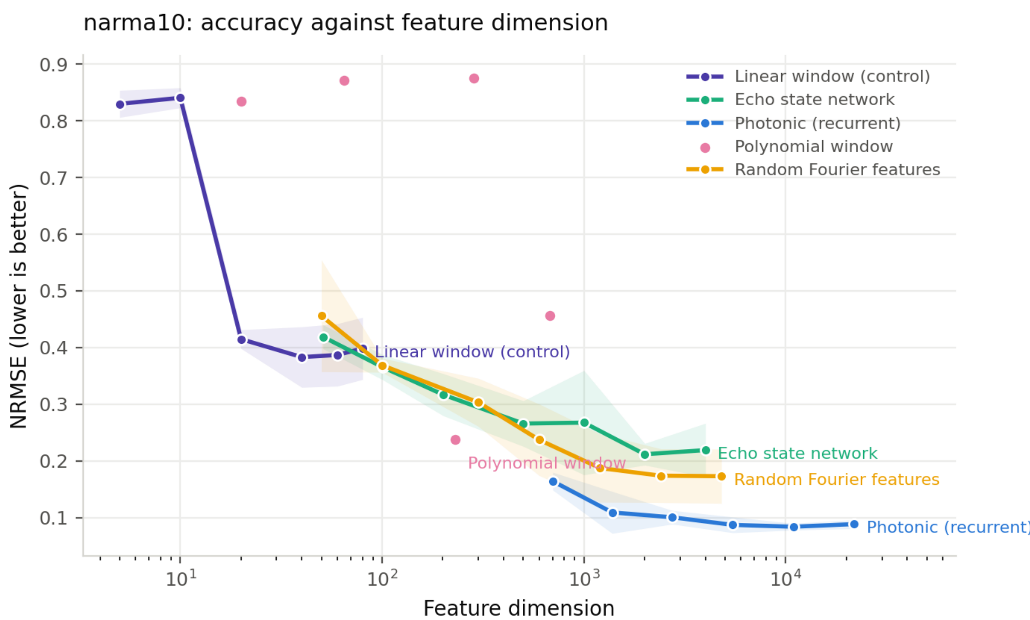

Figure 1: NRMSE against feature dimension on NARMA-10. The echo state network saturates beyond ∼2000 units; the photonic map does not. A single photonic reservoir at 701 features already matches the echo state network's best.

4.3 The encoding window tracks the task order

NARMA-N's target contains $u_t u_{t-N+1}$. A linear-optical map applies a unitary to an encoding vector, so it can only form that product if both lags appear in the same encoding. Sweeping the encoding window with everything else fixed gives optima of 8, 12 and 25 for NARMA-5, NARMA-10 and NARMA-20 respectively — tracking the task order rather than being a free-floating hyperparameter.

We note that in the small hardware-viable configuration used for that sweep, the feedback contributes only 5–11% and is slightly harmful on NARMA-5. The factor of two reported in Section 4.1 is measured at the accuracy-tuned configuration. The benefit of recurrence grows with model capacity, and we report both numbers.

4.4 Memory and information processing capacity

System	Dim.	Linear MC	IPC	IPC/feature
photonic no feedback	132	10.5	111.9	0.85
linear window 20	20	19.1	13.0	0.65
esn 200	201	27.0	123.9	0.62
photonic	132	11.1	69.3	0.52
esn 500	501	29.6	213.9	0.43

Table 4: Linear memory capacity (Jaeger, 2001) and information processing capacity (Dambre et al., 2012) on i.i.d. uniform input, mean over 3 seeds.

The photonic map carries the highest information processing capacity *per feature* but substantially less linear memory than an echo state network. It trades memory depth for nonlinearity, which is consistent with where it wins: NARMA targets are low-order polynomials of a bounded stretch of history.

4.5 Echo state property

Perturbing the initial state and driving two copies with identical input, the perturbation decays for $g_{\text{fb}} \leq 1.0$ and persists indefinitely for $g_{\text{fb}} \geq 1.5$. The memory horizon lengthens with feedback gain — 44, 72, 109 and 352 steps at $g_{\text{fb}} = 0, 0.3, 0.6, 1.0$ — and the best accuracy occurs just below the transition, the standard edge-of-stability picture. All reported experiments use $g_{\text{fb}} = 0.3$, inside the contracting regime.

5 Robustness at measured hardware noise

Noise is simulated with Perceval's device model and threshold detectors, at operating points read from Quandela's cloud API: Ascella at Hong–Ou–Mandel visibility 86.36%, $g^{(2)}(0) = 1.95\%$, transmittance 2.44%, clock 80 MHz; Belenos at 82.7%, 18.2% and clock 4.94 MHz. The configuration tested is the hardware-viable one — two photons in twelve modes — not the accuracy-tuned configuration, which no current device could execute.

Coincidences/step	NRMSE
∞	0.2748 ± 0.0279
30	0.4901 ± 0.0354
100	0.5340 ± 0.0520
300	0.4626 ± 0.0192
1000	0.4113 ± 0.0439
3000	0.4038 ± 0.0480
10000	0.3798 ± 0.0153
30000	0.3433 ± 0.0303

Indistinguishability	NRMSE
0.5	0.2484
0.6	0.2482
0.7	0.2483
0.8	0.2468
0.85	0.2478
0.9	0.2486
0.95	0.2483
1	0.2479

$g^{(2)}(0)$	NRMSE
0	0.2479
0.02	0.2480
0.05	0.2482
0.1	0.2485
0.2	0.2492
0.3	0.2498

Table 5: Sensitivity to finite sampling (left), photon indistinguishability (centre) and multi-photon emission (right), NARMA-10.

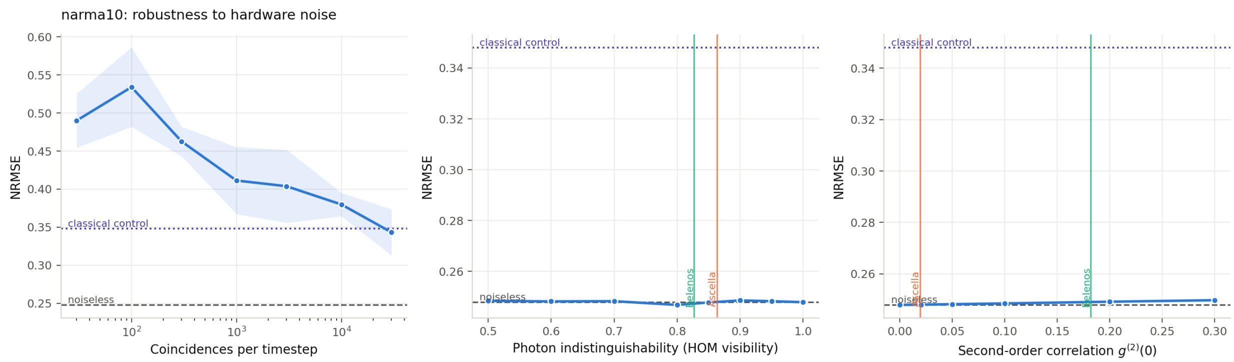

Figure 2: Accuracy against each noise source. Dashed line: noiseless. Dotted line: the classical control. Vertical rules mark the measured Ascella and Belenos operating points, both well inside the flat region.

Device imperfections are close to irrelevant. Accuracy is flat across Hong–Ou–Mandel visibility from 0.50 to 1.00 and rises only 0.8% as $g^{(2)}(0)$ goes from 0 to 0.30. Running the full Ascella and Belenos models — every source at once — gives 0.2483 and 0.2465 against a noiseless 0.2479.

Finite sampling, by contrast, dominates. Extending the sweep to 10^6 coincidences per timestep gives NRMSE $0.418, 0.399, 0.321, 0.305, 0.285, 0.283$ at $10^3, 10^4, 3\cdot10^4, 10^5, 3\cdot10^5, 10^6$ respectively, against 0.275 in the infinite-shot limit and 0.348 for the classical control. Two thresholds follow: roughly $3\cdot10^4$ coincidences per timestep are needed for the reservoir to beat its classical control at all, and roughly $3\cdot10^5$ to come within 4% of its own noiseless limit, beyond which it is converged.

The asymmetry has a simple explanation. A reservoir is a random feature map, and the readout is refitted on whatever map the device happens to realise; a slightly perturbed map is still a perfectly good map, and imperfect indistinguishability merely interpolates towards the distinguishable-particle distribution (Tichy, 2014), which is itself a usable nonlinear feature map. Sampling noise is different in kind: it corrupts every feature independently at every timestep and no readout can absorb it.

5.1 Consequence: optimise rate, not photon quality

Detected n-fold coincidence rate scales as η^n in the transmittance η, so photon number is the dominant lever on the quantity that actually matters.

Platform	n	Coincidences/s	Time for 3×10^4
Ascella	2	4.76×10^4	0.63 s
Ascella	3	1.16×10^3	25.81 s
Ascella	4	2.84×10^1	18 min
Belenos	2	1.16×10^4	2.59 s
Belenos	3	5.60×10^2	53.56 s
Belenos	4	2.71×10^1	18 min

Table 6: Coincidence rate and the time required to collect 3×10^4 events per timestep, at the measured operating points. Reaching the near-converged threshold of 3×10^5 costs ten times as long: 6.3 s per timestep at two photons on Ascella, against 4.3 minutes at three.

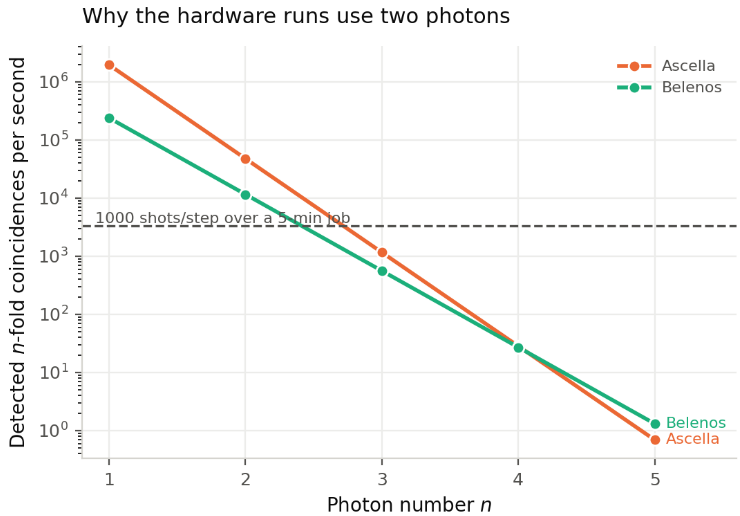

Figure 3: Coincidence rate against photon number.

This inverts the usual instinct that a quantum reservoir needs high-quality single photons. For this class of model, two photons at high rate is strictly preferable to three at low rate. It also explains a failed earlier attempt of our own: a three-photon probe on Belenos returned ∼48 counts from 1000 requested shots spread over 56 Fock bins, and its features correlated only 0.18–0.48 with simulation. That was sampling noise, not a calibration fault.

6 Is it the interference, or a random nonlinear map?

Random Fourier features come close to this model on several tasks, so the question that decides whether "photonic" is load-bearing is not whether a random nonlinear map helps — it plainly does — but whether the *interference* contributes. Partial distinguishability answers it cleanly, because it interpolates exactly between the two hypotheses at otherwise identical circuit, phases, shot count and readout. At $\mathcal{I} = 0$ the photons are classical distinguishable particles and the distribution is the permanent of $|U|^2$, a positive matrix and a classical stochastic feature map. At $\mathcal{I} = 1$ they are indistinguishable bosons and the distribution is $|\mathrm{perm}(U)|^2$, with interference among the $n!$ assignments.

We evaluate in the infinite-shot limit so sampling cannot mask the effect, pair observations across seeds (a seed fixes both the circuit draw and the data), and require an effect to clear both $p < 0.05$ and half a seed standard deviation before calling it an effect.

Task	n	$\mathcal{I}$=0	$\mathcal{I}$=1	ratio	p	verdict
Channel eq.	2	0.1726	0.1728	0.999	0.571	no detectable effect
Channel eq.	3	0.1725	0.1737	0.993	0.541	no detectable effect
Hénon (h=4)	2	0.8811	0.8830	0.998	0.571	no detectable effect
Hénon (h=4)	3	0.8811	0.8849	0.996	0.034	*hurts*
NARMA-10	2	0.2465	0.2373	1.039	0.886	no detectable effect
NARMA-10	3	0.2225	0.2198	1.012	0.280	no detectable effect
Parity (d=3)	2	0.7506	0.7901	0.950	0.324	no detectable effect
Parity (d=3)	3	0.3140	0.3835	0.819	0.166	no detectable effect

Table 7: Effect of photon indistinguishability, median NRMSE over 3 seeds, paired across seeds. Seven of eight configurations show no detectable effect; ratios scatter in both directions and the one significant case is adverse.

We therefore state plainly that **this architecture is a classical random nonlinear feature map realised in optics**. The interferometer and Fock measurement supply a useful high-dimensional map; the indistinguishability of the photons is not what makes it work.

One distinction in Table 7 matters. On the parity task *photon number* helps substantially — $n = 3$ reaches 0.314 against $n = 2$'s 0.751 — while indistinguishability does not. The resource being exploited is Fock-space dimension $\binom{m}{n}$, not interference.

Two consequences follow. Scientifically, this explains Section 5: insensitivity to Hong–Ou–Mandel visibility there and the absence of an interference effect here are the same fact observed twice. Practically, it lowers the hardware requirement, because a source for this application need not be indistinguishable — and HOM visibility, normally the headline figure of merit for a single-photon source, is not the specification to optimise. Transmittance is (Section 7).

7 When would photonics beat simulating it?

The obvious objection to a photonic reservoir at these scales is that a laptop simulates it faster, and it deserves a number rather than a deflection. Classical simulation cost grows combinatorially in photon number: composing the unitary costs $O(dm^3)$ and the permanent batch costs $\binom{m}{n} \cdot n! \cdot n$ multiply-adds per timestep. The device's cost grows as $1/\eta^n$, since it must accumulate a fixed number of coincidences and the rate falls as η^n. Which grows faster settles the question.

Taking the measured throughput of our implementation (7.3×10^7 complex multiply-adds per second, single core) and the measured platform specifications, at the 3×10^4-coincidence threshold established in Section 5:

Platform	n	Coincidences/s	Time for 3×10^4
Ascella	2	4.76×10^4	0.63 s
Ascella	3	1.16×10^3	25.81 s
Ascella	4	2.84×10^1	18 min
Belenos	2	1.16×10^4	2.59 s
Belenos	3	5.60×10^2	53.56 s
Belenos	4	2.71×10^1	18 min

Table 8: Coincidence rate and the time to collect 3×10^4 events per timestep.

There is no crossover at any photon number on current hardware. The device is closest at $n = 2$ and is still 1.6×10^3 times slower than simulating it on one core for Ascella, and 6.6×10^3 times slower for Belenos.

The useful part is that the requirement is now quantitative. A crossover at $n = 12$ in 24 modes would need $\eta \geq 0.105$ on Ascella against its present 0.0244, a factor of four, or $\eta \geq 0.132$ on Belenos, a factor of three. The required transmittance *falls* with photon number — 0.978 at $n = 2$ down to 0.105 at $n = 12$ — because classical cost outgrows the rate penalty. High photon number is therefore the regime where optics could win, provided loss drops far enough to reach it. At $n = 2$ and $n = 3$ on Belenos the required transmittance exceeds unity, meaning simulation is so cheap there that no device could beat it.

This is why we make no computational-advantage claim, and why we read loss reduction rather than photon count as the operative engineering target for this class of model.

8 Hardware execution

No result in this paper is a measurement on a quantum processor. Both `qpu:ascella` and `qpu:belenos` reported `maintenance` throughout this work.

The execution path itself has been run end to end on Quandela's cloud, including chunked submission, threshold-detector sampling, coincidence post-selection and result caching. Over 600 timesteps at 8×10^3 requested coincidences per step, device-measured features correlate with simulation at 0.999 (worst case 0.994). This is the quantitative validation of the two-photon decision: the earlier three-photon probe on the Belenos QPU reached only 0.18–0.48, because at η^3 it collected ~ 48 counts per step across 56 Fock bins.

Quandela's cloud emulator platforms also corroborate Section 5 independently. Submitting 30 identical phase settings to each platform at 2×10^4 shots and measuring total variation against an exact local calculation gives 0.0211 for the noiseless `sim:slos` and 0.0288 for `sim:ascella`, against 0.0193 ± 0.0004 for a perfect multinomial sampler at that shot count — a $+26\sigma$ excess for the emulator. The device model is therefore real but small: it displaces the distribution by 0.008 on top of a sampling error of 0.019. That is the same ordering Perceval's device model gives, reached independently through Quandela's own emulator.

We note that the downstream NRMSE ordering is not evidence on this point in either direction: `sim:ascella` scored better than the noiseless `sim:slos` in the 600-step run, which is seed-to-seed scatter in a regression rather than a statement about the underlying distributions. Only the distribution comparison settles it.

Because a closed feedback loop cannot be batched into a single cloud job (Section 3.2), two protocols are provided. In *replay*, the recurrence is evaluated in simulation to produce the phase trajectory, which is then replayed on the device as one batched job and the readout trained on device-measured features; this measures the device's feature map faithfully but the feedback path is simulated, and we report it as such. In *open-loop*, feedback is disabled and every timestep is independent, so the run is end-to-end on the device at the cost of a weaker model.

In practice the open-loop protocol is both the stronger claim and the better result: over 600 timesteps it reaches NRMSE 0.347 against 0.393 for replay and 0.423 for the classical control, a lift of 1.22. This is consistent with Section 4.3, where the feedback contributes little in the small hardware-viable configuration. Giving it up therefore costs almost nothing at the scale a current device can run, while removing the only caveat the replay protocol carries — a useful result for anyone planning a hardware demonstration.

Two operational findings are worth recording for others attempting this. On the free tier a job is cancelled at five minutes: one of our submissions was terminated at 307 s having completed 4% of its iterations, which forces chunked submission with per-chunk caching. And the shot budget per call must be set explicitly and generously, since the post-selection that follows consumes it.

9 Limitations

Everything here is simulated. We claim no quantum advantage. The comparison is between feature maps on classical data, and a passive interferometer is a classically simulable device at these photon numbers by construction.

Interference does not contribute. Section 6 tests this directly and finds no detectable effect on seven of eight configurations. Readers should treat this as a result about the architecture rather than a caveat: what the optics supplies is a high-dimensional random nonlinear map, and the useful resource is Fock-space dimension.

No computational crossover. Section 7 shows the device is at best 1.6×10^3 times slower than simulating it on one core, at any photon number, on current hardware.

A negative result on the financial task. On S&P 500 realised volatility the model does not win and nothing is statistically distinguishable. We include the task because it was pre-registered, and we draw no conclusion from it.

A negative result on the predecessor architecture. An earlier version of this work used the same optics without recurrence, encoding a sliding window directly as phases. It did not outperform a ridge regression on the raw window (MSE $5.38 \cdot 10^{-3}$ against $5.24 \cdot 10^{-3}$ on NARMA-10), and no setting of encoding gain, circuit depth, photon number or mode count changed that. The cause was an encoding that spanned only 0.58 radians of the available 2π, leaving the optical features with $16\times$ less variance than the classical block beside them. We report this because the failure mode is easy to reproduce and hard to detect without a classical control in the table.

Feature dimension. The tuned configuration uses more features than the baselines. Section 4.2 is the honest version of Table 1, and any claim should be read against it.

Statistical power. With 200–600 test points the Model Confidence Set does not discriminate. Our conclusions rest on Diebold–Mariano tests.

10 Conclusion

Feeding a leaky-integrated state back into the phase encoding of a fixed interferometer turns a windowed optical feature map into a reservoir, and does so at no additional photonic cost, since the feedback is classical and happens between measurement batches. The resulting model outperforms tuned classical feature maps on NARMA-10 and NARMA-20 at matched feature dimension, ties on Mackey–Glass, and does not win on realised volatility.

For hardware, the operative result is that the model is insensitive to the imperfections photonic platforms are usually characterised by, and limited instead by how many coincidences can be collected. That converts a vague question about device quality into a concrete engineering target, and suggests that the useful direction for this class of model is fewer photons at higher rate.

References

A. Amanov and K. Azamov. Hybrid photonic quantum reservoir computing for high-dimensional financial surface prediction. *arXiv preprint arXiv:2603.10707*, 2026.

Donald W. K. Andrews. Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica*, 59(3):817–858, 1991.

P. Bienstman et al. Passive photonic reservoir computing in a nalm cavity. *Nature Communications*, 2025.

A. Branco et al. Hard to beat: The overlooked impact of rolling windows in the era of machine learning. *Journal of Forecasting*, 43(5):1247–1273, 2024.

Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196, 2009.

Joni Dambre, David Verstraeten, Benjamin Schrauwen, and Serge Massar. Information processing capacity of dynamical systems. *Scientific Reports*, 2:514, 2012.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 1995.

Jorge García-Beni et al. Scalable photonic platform for quantum reservoir computing. *Physical Review Applied*, 20(1):014024, 2023.

Peter Reinhard Hansen, Asger Lunde, and James M. Nason. The model confidence set. *Econometrica*, 79(2):453–497, 2011.

N. Heurtel et al. Perceval: a software platform for discrete variable photonic quantum computing. *Quantum*, 2023.

Herbert Jaeger. The "echo state" approach to analysing and training recurrent neural networks. *GMD Report 148, German National Research Center for Information Technology*, 2001.

X. Li, R. Mukhopadhyay, A. Bayat, and A. Habibnia. Quantum reservoir computing for realized volatility forecasting. *arXiv preprint arXiv:2505.13933*, 2025.

Wolfgang Maass, Thomas Natschläger, and Henry Markram. Real-time computing without stable states: A new framework for neural computation based on perturbations. *Neural Computation*, 14(11):2531–2560, 2002.

N. Maring et al. A versatile single-photon-based quantum computing platform. *Nature Photonics*, 2024.

S. Nerenberg et al. Photon-quarc: photon-number-resolving quantum reservoir computing in linear optics. *Optica*, 2025.

Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.

Dimitris N. Politis and Joseph P. Romano. The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994.

Ali Rahimi and Benjamin Recht. Random features for large-scale kernel machines. In *Advances in Neural Information Processing Systems*, 2007.

Malte C. Tichy. Interference of identical particles from entanglement to boson-sampling. *Journal of Physics B*, 2014.